

Invertible Video Signal Processing

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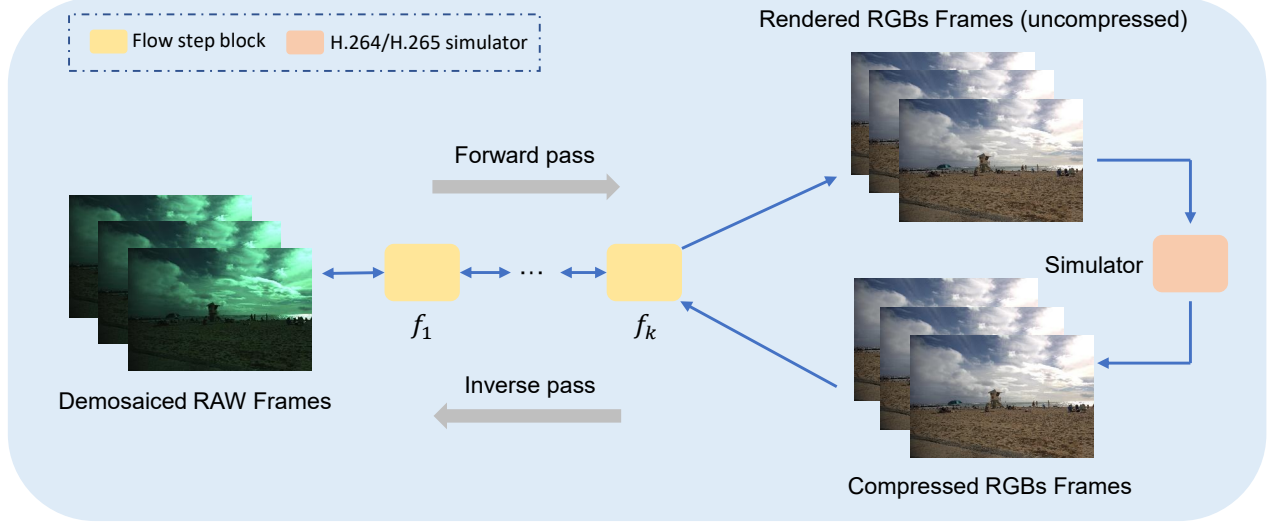


Figure 1: Our invertible video signal processing framework for RGB video rendering and reconstruction. Our invertible video pipeline needs to consider both the reconstruction quality and temporal consistency. Thus, for the forward pass, we sequentially feed W frames to the network to obtain the RGB frames. We also propose a more proper compression simulator for video codecs like H.264 and H.265. During training, in the reverse process, we replace the not-centered frames with the ground-truth RGB images to maintain the symmetric property of the sliding window.

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1 Introduction

Raw sensor data plays a vital role in photography and computer vision, as it preserves linear, high-bit-depth scene irradiance before information is lost through the camera image signal processing (ISP) pipeline. While recent research has explored learning-based methods to reconstruct RAW images from compressed RGB data [Xing et al. 2021], little attention has been paid to the video domain. This gap is significant: modern digital cameras typically discard RAW videos due to prohibitive storage and transmission costs, leaving

only compressed RGB videos available for downstream processing. However, many applications—such as HDR video, video editing, and computational photography—would benefit from access to temporally consistent RAW video sequences.

In this work, we present the first framework for RAW video reconstruction by extending invertible ISP modeling to the temporal domain. Our approach leverages the reversibility of normalizing flow models [Dinh et al. 2017] to jointly support high-quality RGB video rendering and RAW video recovery, while preserving temporal consistency between frames. To address challenges unique to video data, we design modules that explicitly account for inter-frame dependencies and integrate differentiable video codec simulation to mitigate the effects of lossy compression. Furthermore, we contribute a new RAW video dataset captured with a Sony RX100M VI, covering various scenes with paired RAW-RGB sequences. To our knowledge, this is the first work for accurate RAW video reconstruction from compressed video files, opening the door to novel applications in video enhancement and computational photography.

*Both authors contributed equally to this research.

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Table 1: Quantitative evaluation between our model and the baseline InvISP [Xing et al. 2021] and Naive CNS [Hu et al. 2020]. The perceptual metrics show that our proposed video signal processing model outperforms InvISP and the other video compression simulation method CNS. Average PSNR is calculated with the average of RGB PSNR and RAW PSNR.

Method	CRF 23				CRF 35				CRF 41			
	RGB		RAW	Average	RGB		RAW	Average	RGB		RAW	Average
	PSNR	SSIM	PSNR	PSNR	PSNR	SSIM	PSNR	PSNR	PSNR	SSIM	PSNR	PSNR
InvISP [Xing et al. 2021]	35.62	0.842	40.76	38.19	31.97	0.761	34.12	33.05	29.98	0.700	25.63	27.81
Naive CNS [Hu et al. 2020]	35.15	0.824	41.71	38.43	32.21	0.740	35.33	33.37	30.11	0.698	32.74	31.43
Ours	35.74	0.837	41.50	38.62	32.70	0.763	36.79	34.75	30.23	0.701	33.40	31.82

2 Method

2.1 Invertible Video Signal Processing Pipeline

Given a RAW video sequence, our objective is to develop a normalizing flow pipeline [Dinh et al. 2017] capable of rendering it into a compressed RGB video and subsequently reconstructing the original RAW data. To ensure temporal consistency across frames, the model processes a sliding window of W consecutive RAW frames for the forward rendering process and W consecutive RGB frames for the inverse reconstruction process, as illustrated in Figure 1. This approach encourages the network to learn spatiotemporal features from adjacent frames. Crucially, our design optimizes only the center frame within each window. During training, the loss is computed solely on the center RGB and center RAW frames; the surrounding frames are replaced with their ground-truth counterparts to stabilize learning. During inference, these surrounding frames are populated with the model’s previous reconstructions. The framework is trained under a bi-directional supervision scheme to jointly optimize both transformation directions as:

$$L_{video} = \sum_{t=1}^T ||f(\mathbf{x}_t) - \mathbf{y}_t||_1 + \lambda ||f^{-1}(\mathbf{y}_t) - \mathbf{x}_t||_1, \quad (1)$$

where $\mathbf{x}_t$ is the input center RAW frame, $\mathbf{y}_t$ is the target center RGB frame and f denotes the learned mapping from RAW data space to sRGB data space.

2.2 Simulator for video codecs compression

To account for the effects of lossy video codecs like H.265/HEVC [Sullivan et al. 2012] in real-world applications, we designed a compression-quality-aware video simulator. Our framework decomposes the simulation into two distinct components: inter-frame and intra-frame compression. The selection of I, P, and B frames is inherently complex, depending on both frame content and the specific codec’s algorithm. To simplify this, we implement a randomized alternation between inter and intra-frame simulation modes. For intra-frame simulation, we adapt the naive CNS module [Hu et al. 2020] by modulating the compression ratio via discrete cosine transform (DCT) tables. For inter-frame simulation, we introduce random motion jittering to approximate the optical flow estimation process in inter-prediction. Specifically, for each macroblock, we sample a dominant motion vector from the range $[0, M]$, where M is a hyperparameter defining the maximum motion magnitude. This vector is then perturbed by a small random jitter within $[0, 0.1]$.

The macroblock is subsequently warped according to this final motion vector to simulate the result of inter-frame prediction. Finally, compression is applied to the resulting residual error.

2.3 Our Captured RAW Video Datasets

Due to the absence of a large-scale public RAW video dataset, we introduce a new dataset comprising 160 video sequences captured from diverse scenes using Sony RX100 VI cameras. Each sequence contains 57 to 114 frames with a resolution of 2748×1936 , recorded at 24 fps in burst mode. We randomly partitioned the dataset into training and test sets, resulting in 140 videos for training and 20 for testing. The LibRaw library was utilized to process the RAW video frames and render the corresponding sRGB video frames.

3 Experiment Results

We evaluate our invertible video signal processing on RAW video reconstruction from compressed RGB videos, as shown in Tab. 1 and Fig. 2. Compared to the per-frame image flow baseline [Xing et al. 2021], our video flow framework achieves more accurate RAW recovery while preserving RGB quality, especially on edges and saturated regions. Incorporating our codec-aware simulation further improves robustness to real-world video compression, outperforming JPEG- and CNS-based alternatives. The method also generalizes well to H.265/HEVC, demonstrating that explicitly modeling inter-frame dependencies and compression artifacts is crucial for temporally consistent and high-quality RAW video reconstruction.

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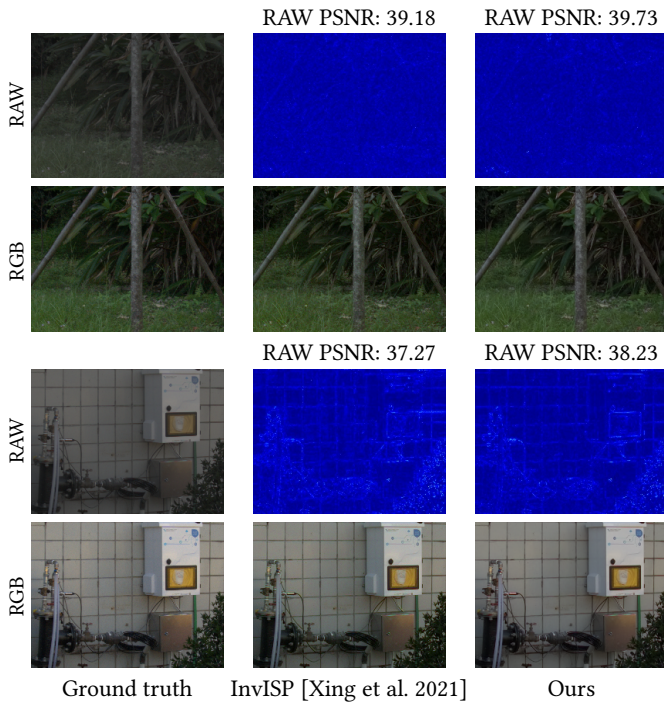


Figure 2: Comparison between image flow [Xing et al. 2021] and our video flow methods on invertible video signal processing task. Error maps are used to visualize.

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